Movie Recommendation System - Final Project Report

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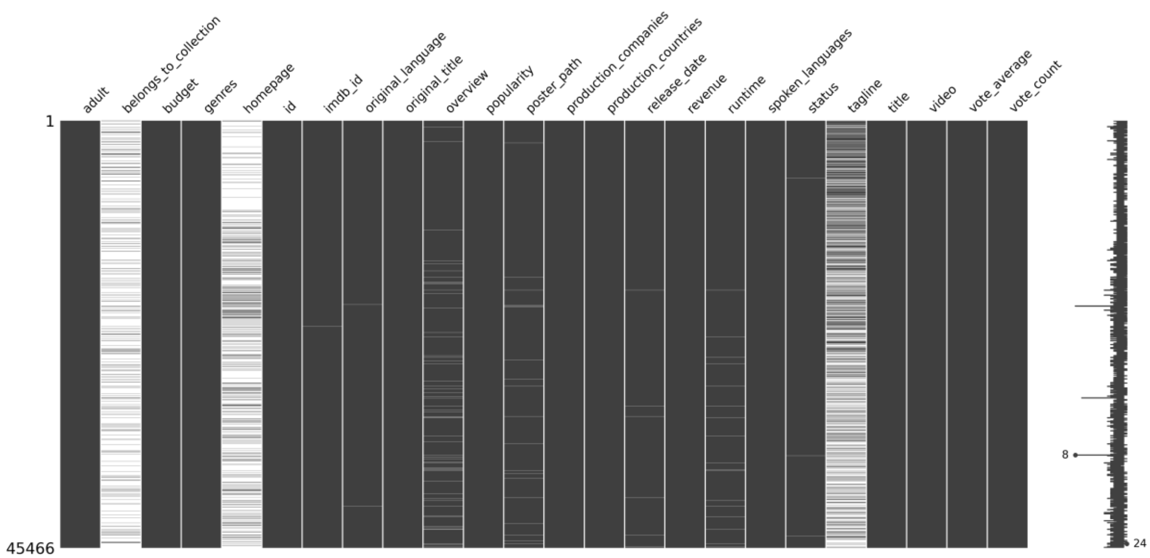
1. Introduction

This report presents the development of a sophisticated Movie Recommendation System as part of the final project for the Machine Learning course. The goal of this project was to build a recommender system that effectively combines multiple approaches including popularity-based, content-based, collaborative filtering, and hybrid methods. To achieve this, multiple datasets were merged, preprocessed, analyzed, and modeled.



1 – missing values

```
merged_df = pd.merge(df, credits_df, on='id', how='inner')
merged_df = pd.merge(merged_df, keywords_df, on='id', how='inner')
merged_df = pd.merge(merged_df, links_df, left_on='id', right_on='tmdbId', how='inner')
merged_df = pd.merge(merged_df, ratings_df, on='movieId', how='inner')

print("Shape of the merged dataframe:", merged_df.shape)
display(merged_df.head().transpose())
```

```
Shape of the merged dataframe: (99810, 30)
```

	0	1	2	3	
adult	False	False	False	False	
budget	30000000	30000000	30000000	30000000	
genres	Animation, Comedy, Family	Animation, Comedy, Family	Animation, Comedy, Family	Animation, Comedy, Family	Ar
id	862	862	862	862	
imdb_id	tt0114709	tt0114709	tt0114709	tt0114709	
original_language	en	en	en	en	
original_title	Toy Story	Toy Story	Toy Story	Toy Story	
overview	Led by Woody, Andy's toys live happily in his ...	Led by Woody, Andy's toys live happily in his ...	Led by Woody, Andy's toys live happily in his ...	Led by Woody, Andy's toys live happily in his ...	Led by Wood
popularity	21.946943	21.946943	21.946943	21.946943	
poster_path	/rhIRbceoE9lR4veEXuwCC2wARtG.jpg	/rhIRbceoE9lR4veEXuwCC2wARtG.jpg	/rhIRbceoE9lR4veEXuwCC2wARtG.jpg	/rhIRbceoE9lR4veEXuwCC2wARtG.jpg	/rhIRbceoE9lR
production_companies	Pixar Animation Studios	Pixar Animation Studios	Pixar Animation Studios	Pixar Animation Studios	
production_countries	United States of America	United States of America	United States of America	United States of America	
release_date	1995-10-30	1995-10-30	1995-10-30	1995-10-30	
revenue	373554033.0	373554033.0	373554033.0	373554033.0	
runtime	81.0	81.0	81.0	81.0	
spoken_languages	English	English	English	English	
status	Released	Released	Released	Released	
title	Toy Story	Toy Story	Toy Story	Toy Story	
video	False	False	False	False	
vote_average	7.7	7.7	7.7	7.7	
vote_count	5415.0	5415.0	5415.0	5415.0	
cast	Tom Hanks, Tim Allen, Don Rickles, Jim Varney,...	Tom Hanks, Tim Allen, Don Rickles, Jim Varney,...	Tom Hanks, Tim Allen, Don Rickles, Jim Varney,...	Tom Hanks, Tim Allen, Don Rickles, Jim Varney,...	Tom Hank
crew	John Lasseter, Joss Whedon, Andrew Stanton, Jo...	John Lasseter, Joss Whedon, Andrew Stanton, Jo...	John Lasseter, Joss Whedon, Andrew Stanton, Jo...	John Lasseter, Joss Whedon, Andrew Stanton, Jo...	John Lasset
keywords	jealousy, toy, boy, friendship, friends, rival...	jealousy, toy, boy, friendship, friends, rival...	jealousy, toy, boy, friendship, friends, rival...	jealousy, toy, boy, friendship, friends, rival...	jealousy, to
movieId	1	1	1	1	
imdbId	114709	114709	114709	114709	

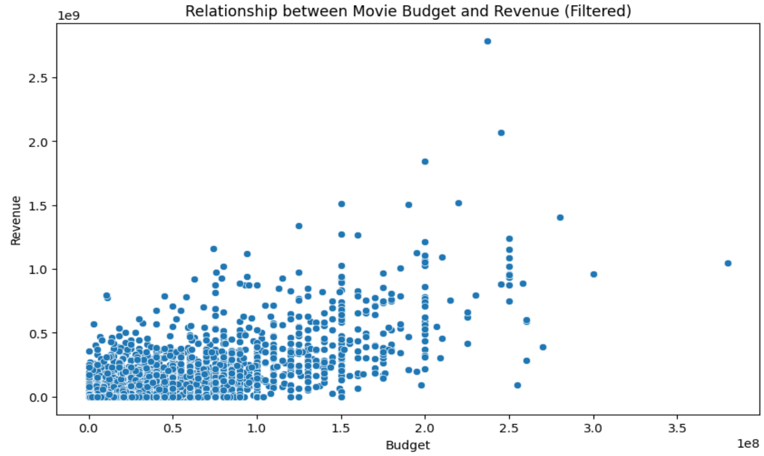
2- merging

2. Data Preprocessing

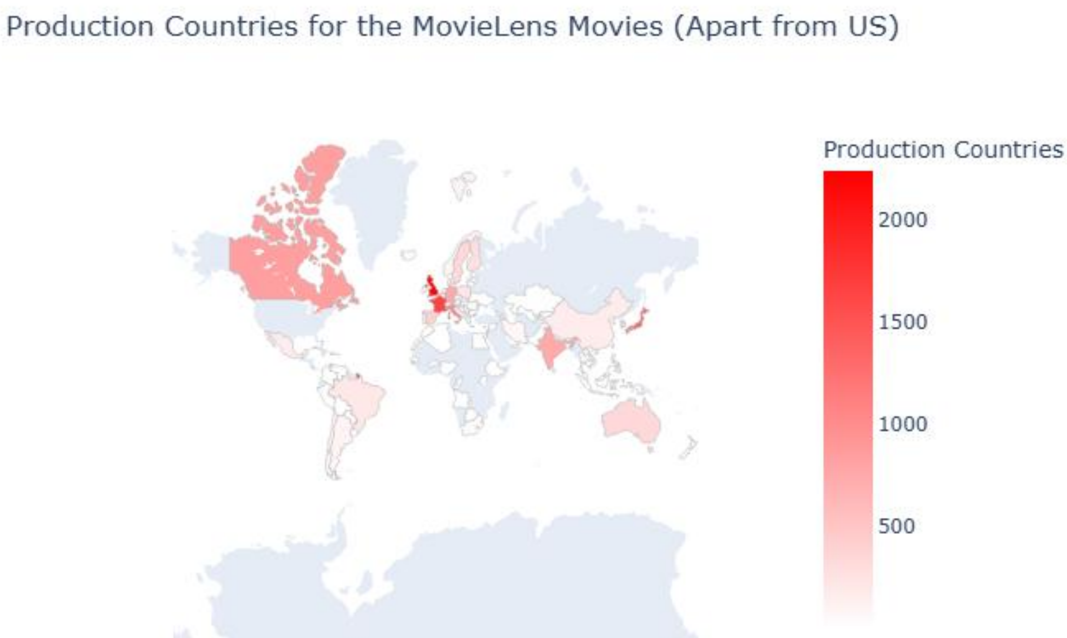
The datasets used in this project include:
- Metadata
- Credits
- Keywords
- Links
- Ratings

The preprocessing steps involved the following:
- Handling missing values and imputing them appropriately
- Parsing and decoding JSON fields into structured rows
- Detecting and removing duplicate records
- Cleaning and normalizing text fields

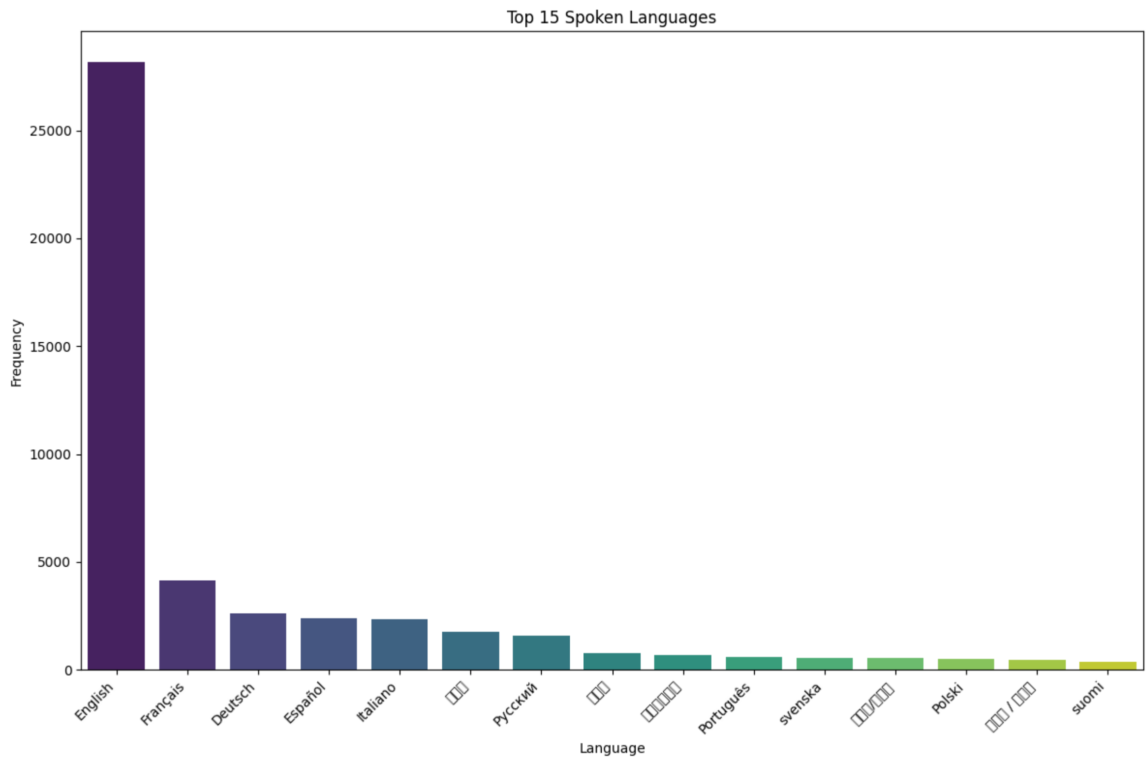
After preprocessing each dataset individually, they were merged into a single comprehensive dataframe.



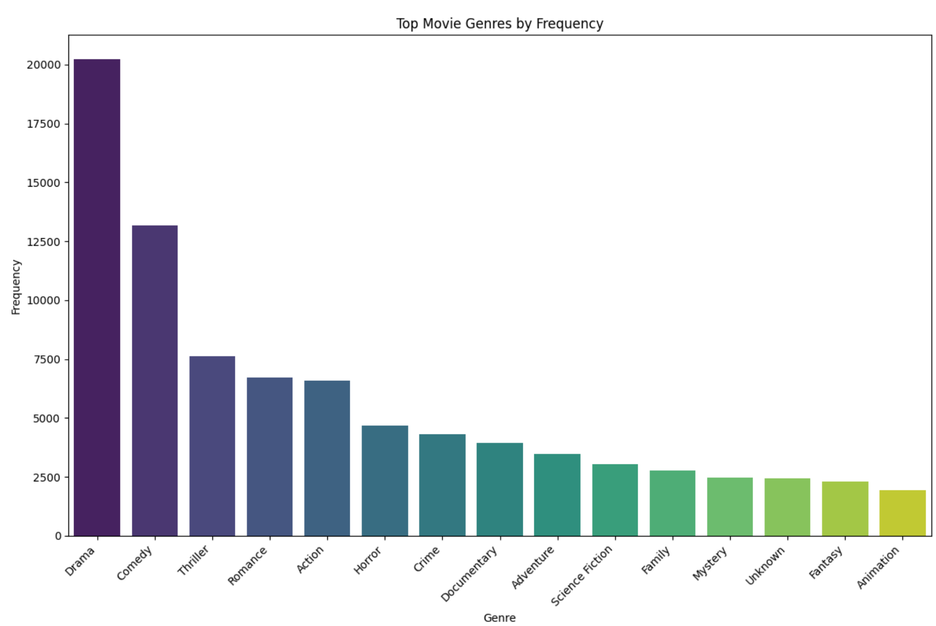
3 – budget revenue correlation



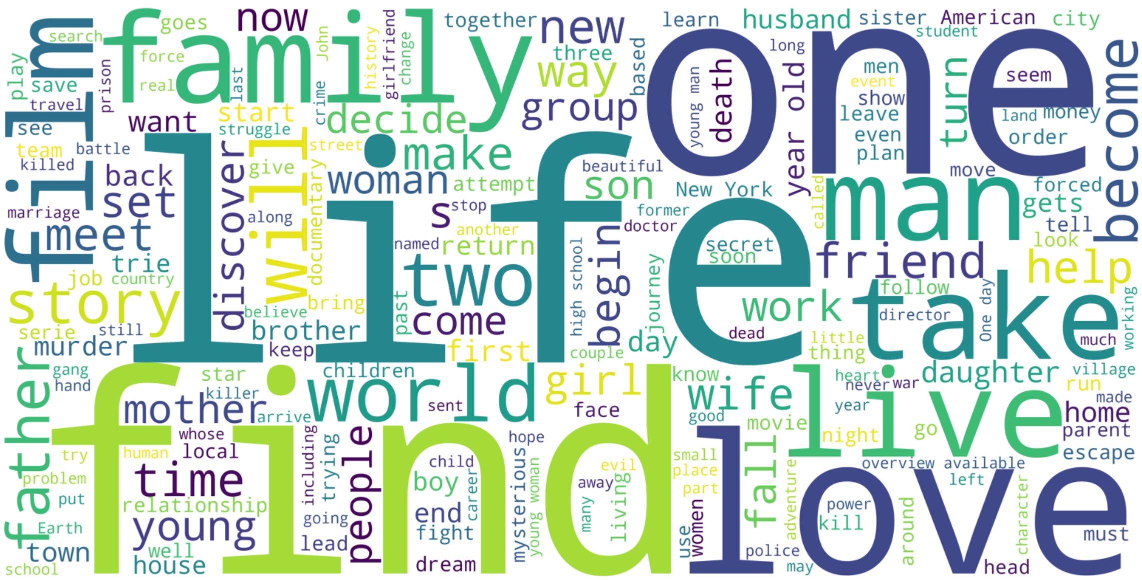
4 – movie production - US



5- language distribution

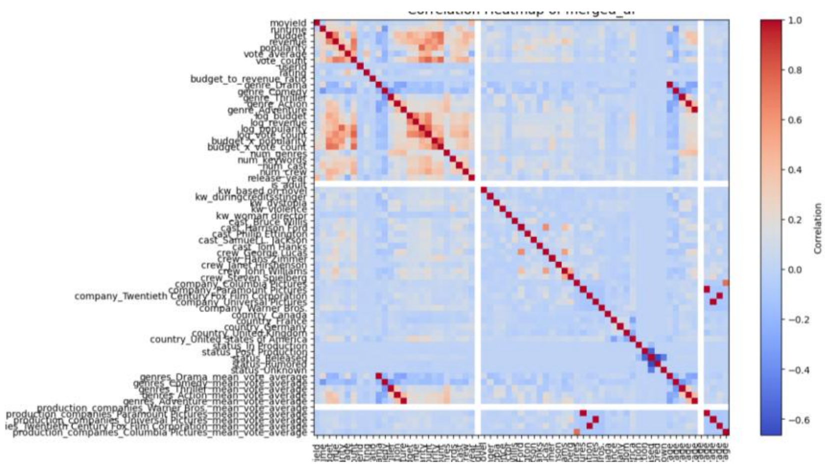


6 – genres frequency



7 - keywords

8 – keywords



9- CORLATION HEATMAP

3. Exploratory Data Analysis (EDA)

A detailed EDA was conducted to understand the structure and distribution of the data. Key insights include:

- A strong positive correlation exists between movie budget and revenue.
- Distribution of movies by decade:
  * 2010s: highest
  * 2000s: second
  * 1990s: third
- Popularity distribution decreases gradually with a long-tail effect.
- Rating distribution approximates a normal distribution.
- Runtime distribution: Most movies range between 90–200 minutes, with the majority between 1–2 hours.
- Top genres (in order): Drama, Comedy, Thriller, Romance, Action, Horror, Crime, Documentary.
- Top languages (in order): English, French, Dutch, Spanish, Italian.
- Top production companies: Warner Bros., Metro-Goldwyn-Mayer, Paramount Pictures, Universal Pictures, 20th Century Fox Film Corporation.
- Top production countries: United States, United Kingdom, France, Germany, Italy, Canada.
- Word cloud analysis of the 'Overview' field highlighted frequent words such as 'life', 'family', 'feel', 'love', and 'find'.

4. Feature Engineering

Several feature engineering techniques were applied to enrich the dataset:
- TF-IDF vectorization for textual features.
- One-hot encoding for categorical variables.
- Creation of new features such as top cast members, directors, production companies, and countries.
- Dimensionality reduction using Singular Value Decomposition (SVD) with 120 components.

```
Mean rating across all movies (C): 6.87
Minimum number of votes (m) for top list consideration (90th percentile): 4183.00

Number of movies qualified for weighted rating calculation: 9972

Top 10 Popular Movies based on Weighted Rating:
```

	title	vote_count	vote_average	weighted_rating
0	Toy Story	5415.0	7.7	7.338731
1	Se7en	5915.0	8.1	7.590923
2	Star Wars	6778.0	8.1	7.631005
3	Leon: The Professional	4293.0	8.2	7.544153
4	Pulp Fiction	8670.0	8.3	7.834952
5	The Shawshank Redemption	8358.0	8.5	7.956674
6	Forrest Gump	8147.0	8.2	7.749152
7	The Lion King	5520.0	8.0	7.513310
8	Jurassic Park	4956.0	7.6	7.266358
9	Schindler's List	4436.0	8.3	7.606502

9- Baseline

```
Content-Based Recommendations for User 1:
```

	movieId	title	vote_average	vote_count
0	6721	Once Upon a Time in China	7.4	99.0
1	158956	Kill Command	5.2	203.0
2	393	Street Fighter	4.1	330.0
3	4565	American Ninja	5.5	71.0
4	4442	The Last Dragon	6.4	71.0
5	59103	The Forbidden Kingdom	6.3	476.0
6	26012	Samurai III: Duel at Ganryu Island	7.4	24.0
7	79695	The Expendables	6.0	2977.0
8	42721	BloodRayne	3.6	120.0
9	704	The Quest	5.3	126.0

```
User 9999 is a cold-start user. Falling back to popularity baseline.

Content-Based Recommendations for Cold-Start User 9999:
```

	title	vote_count	vote_average	weighted_rating	movieId
0	Toy Story	5415.0	7.7	7.338731	1
1	Se7en	5915.0	8.1	7.590923	47
2	Star Wars	6778.0	8.1	7.631005	260
3	Leon: The Professional	4293.0	8.2	7.544153	293
4	Pulp Fiction	8670.0	8.3	7.834952	296
5	The Shawshank Redemption	8358.0	8.5	7.956674	318
6	Forrest Gump	8147.0	8.2	7.749152	356
7	The Lion King	5520.0	8.0	7.513310	364
8	Jurassic Park	4956.0	7.6	7.266358	480
9	Schindler's List	4436.0	8.3	7.606502	527

10 – CB (content – based)

User-based k-NN recommendations for User 1:

	movieId	title	predicted_rating
0	183	Mute Witness	5
1	309	Red Firecracker, Green Firecracker	5
2	702	Faces	5
3	759	Maya Lin: A Strong Clear Vision	5
4	764	Heavy	5
5	845	The Day the Sun Turned Cold	5
6	893	Mother Night	5
7	984	The Pompatus of Love	5
8	1056	Jude	5
9	1312	Female Perversions	5

Matrix Factorization (SVD) recommendations for User 1:

	movieId	title	predicted_rating
0	318	The Shawshank Redemption	3.686334
1	969	The African Queen	3.632321
2	3462	Modern Times	3.586802
3	858	The Godfather	3.580284
4	1228	Raging Bull	3.578779
5	1212	The Third Man	3.535737
6	898	The Philadelphia Story	3.528758
7	2571	The Matrix	3.518022
8	1221	The Godfather: Part II	3.511670
9	2542	Lock, Stock and Two Smoking Barrels	3.511210

11 – CF (Collaborative Filtring)

5. Modeling

Multiple recommendation models were implemented and evaluated:

1. **Popularity Baseline**:
   - Recommends the most popular movies overall.
   - Example recommendations: Toy Story, Jumanji, Heat, GoldenEye, Casino.

2. **Content-Based Filtering**:
   - Suggests movies based on the similarity of content features to user preferences.

3. **Collaborative Filtering**:
   - Matrix factorization methods were used.
   - Achieved RMSE: 0.90.
   - User-based KNN achieved RMSE: 0.96.

4. **Hybrid Model**:
   - Combines content-based and collaborative filtering approaches.
   - Collaborative filtering coefficient converged towards 1, while content-based coefficient converged towards 0.
   - Achieved RMSE: 0.89–0.70.
   - Alternative hybrid configuration achieved RMSE: 1.06.

6. Conclusion

This project successfully developed a robust movie recommendation system through a structured pipeline of data preprocessing, exploratory analysis, feature engineering, and model implementation. Among the tested approaches, the hybrid model provided the most promising results, demonstrating the power of combining content-based and collaborative filtering techniques. The achieved RMSE scores highlight the model's strong predictive performance. Future work may include integrating deep learning approaches to further improve recommendation quality.